

Future Stories (FSCE)-Style Sample Paper — Adventure & Beacon Maths

Time allowed: 45 minutes**Total marks:** 40**Questions:** 40**Calculator:** Not allowed

Name: _____

Date: _____

Instructions

Answer every question. Tick the box next to the answer you think is correct.

The number in square brackets [] shows the marks for each question.

Show your working in the boxes provided. You may not use a calculator.

Diagrams are not always drawn to scale.

- 1 What number does the Roman numeral XLVII represent? [1]

☐ A 47☐ B 57☐ C 43☐ D 52

- 2 One January morning, four locations record these temperatures: Riverside -6°C , Hilltop -2°C , Valley 3°C , Lakeside -9°C . Which location recorded the coldest temperature? [1]

MORNING TEMPERATURES

☐ A Lakeside☐ B Riverside☐ C Hilltop☐ D Valley

- 3 Round 68,472 to the nearest thousand. [1]

ROUNDING TO THE NEAREST 1,000

☐ A 68000☐ B 69000☐ C 68500☐ D 68400

- 4 In the number 803,516, what is the value of the digit 5? [1]

PLACE VALUE OF 803,516

100,000s	10,000s	1,000s	100s	10s	1s
8	0	3	5	1	6

- ☐ A 500 ☐ B 5000
☐ C 50 ☐ D 5
- 5 At 6am the temperature was -4°C . By midday it had risen by 11°C . What was the temperature at midday? [1]
☐ A 7°C ☐ B 15°C
☐ C -7°C ☐ D 6°C
- 6 A school gate has the founding year carved into the stonework as MCMXLVIII. In which year was the school founded? [1]
☐ A 1948 ☐ B 1947
☐ C 1958 ☐ D 1848

Working

- 7 Work out 236×34 . [1]
☐ A 8024 ☐ B 7994
☐ C 8124 ☐ D 8004
- 8 A youth club has 187 pupils to transport in minibuses that each hold 12 pupils. What is the smallest number of minibuses needed so that every pupil has a seat? [1]
☐ A 16 ☐ B 15
☐ C 17 ☐ D 14

Working

- 9 Work out $8 + 6 \times (9 - 4)$. [1]
☐ A 38 ☐ B 70
☐ C 34 ☐ D 42

- 10 Three classes collected paper for recycling over a week, as shown in the table. The school's target was 400 kg. By how many kilograms did the three classes miss the target? [1]

PAPER COLLECTED THIS WEEK

Class	Paper collected
Class A	128 kg
Class B	96 kg
Class C	154 kg

- ☐ A 22 kg ☐ B 18 kg
☐ C 26 kg ☐ D 32 kg

Working

- 11 Which of these numbers is a factor of 84? [1]

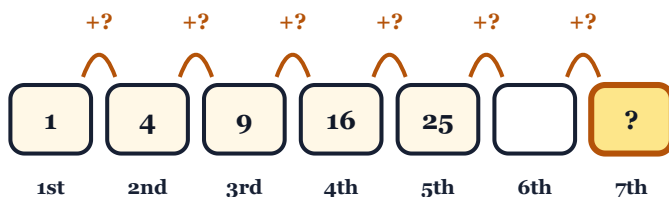
- ☐ A 14 ☐ B 9
☐ C 16 ☐ D 25

- 12 Which of these numbers is a prime number? [1]

- ☐ A 61 ☐ B 51
☐ C 57 ☐ D 63

- 13 The sequence of square numbers begins 1, 4, 9, 16, 25, ... What is the 7th square number? [1]

SQUARE NUMBERS



- ☐ A 49 ☐ B 36
☐ C 42 ☐ D 64

- 14 What is the value of 5^3 (5 cubed)? [1]

- ☐ A 125 ☐ B 15
☐ C 25 ☐ D 10

- 15 Which fraction is equivalent to $\frac{3}{4}$?

[1]

THE FRACTION $\frac{3}{4}$



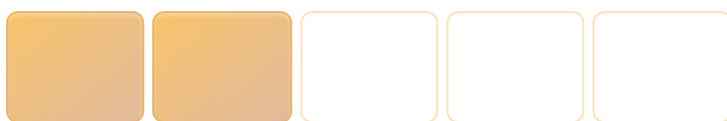
4 equal parts

- ☐ A $\frac{9}{12}$ ☐ B $\frac{6}{9}$
☐ C $\frac{8}{10}$ ☐ D $\frac{5}{8}$

- 16 A charity fun run is 15 km long. A runner has completed $\frac{2}{5}$ of the distance. How many kilometres has she run?

[1]

THE FRACTION $\frac{2}{5}$



5 equal parts

- ☐ A 6 km ☐ B 9 km
☐ C 3 km ☐ D 10 km

- 17 Convert $\frac{7}{20}$ to a decimal.

[1]

- ☐ A 0.35 ☐ B 0.7
☐ C 0.135 ☐ D 3.5

- 18 A stadium has 25% of its 3,200 seats reserved for members. How many seats are reserved for members?

[1]

- ☐ A 800 ☐ B 750
☐ C 850 ☐ D 900

- 19 Which of these is the smallest: $\frac{3}{8}$, $\frac{2}{5}$, $\frac{1}{3}$, or 0.3?

[1]

- ☐ A 0.3 ☐ B $\frac{3}{8}$
☐ C $\frac{2}{5}$ ☐ D $\frac{1}{3}$

Working

- 20 Work out $\frac{1}{4} + \frac{1}{6}$, giving your answer in its simplest form.

[1]

- ☐ A $\frac{5}{12}$ ☐ B $\frac{2}{10}$
☐ C $\frac{1}{5}$ ☐ D $\frac{7}{12}$

- 21 A bike originally costs £120. In a sale it is reduced by 15%. What is the sale price?

[1]

- ☐ A £102 ☐ B £105
☐ C £18 ☐ D £96

Working

22 Round 4.267 to 1 decimal place. [1]

- | | |
|---|--|
| ☐ A 4.3
☐ C 4.27 | ☐ B 4.2
☐ D 4.1 |
|---|--|

23 Simplify the ratio 18 : 24 to its simplest form. [1]

- | | |
|---|--|
| ☐ A 3 : 4
☐ C 9 : 12 | ☐ B 6 : 8
☐ D 2 : 3 |
|---|--|

24 Joe and Priya share £360 in the ratio 4 : 5. How much does Priya, who has the larger share, receive? [1]

JOE : PRIYA	
Joe	
Priya	

- | | |
|--|--|
| ☐ A £200
☐ C £180 | ☐ B £160
☐ D £220 |
|--|--|

25 A map has a scale of 1 cm : 5 km. The distance between two villages on the map is 6.4 cm. What is the real distance between the villages? [1]

- | | |
|--|--|
| ☐ A 32 km
☐ C 11.4 km | ☐ B 30 km
☐ D 64 km |
|--|--|

26 A pancake recipe for 4 people uses 200 g of flour. How much flour is needed to make pancakes for 10 people? [1]

- | | |
|--|--|
| ☐ A 500 g
☐ C 400 g | ☐ B 450 g
☐ D 550 g |
|--|--|

27 In a school choir, the ratio of sopranos to altos is 3 : 2, and the ratio of altos to basses is 4 : 3. If there are 8 altos, how many basses are there? [1]

- | | |
|--|---|
| ☐ A 6
☐ C 4 | ☐ B 8
☐ D 10 |
|--|---|

Working

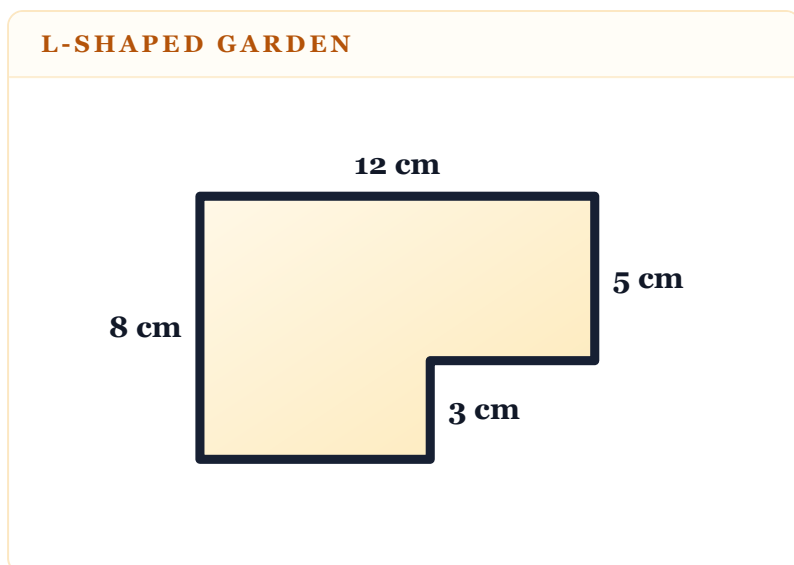
28 Three angles lie on a straight line. Two of them measure 55° and 68° . What is the size of the third angle? [1]

- | | |
|--|--|
| ☐ A 57°
☐ C 62° | ☐ B 65°
☐ D 50° |
|--|--|

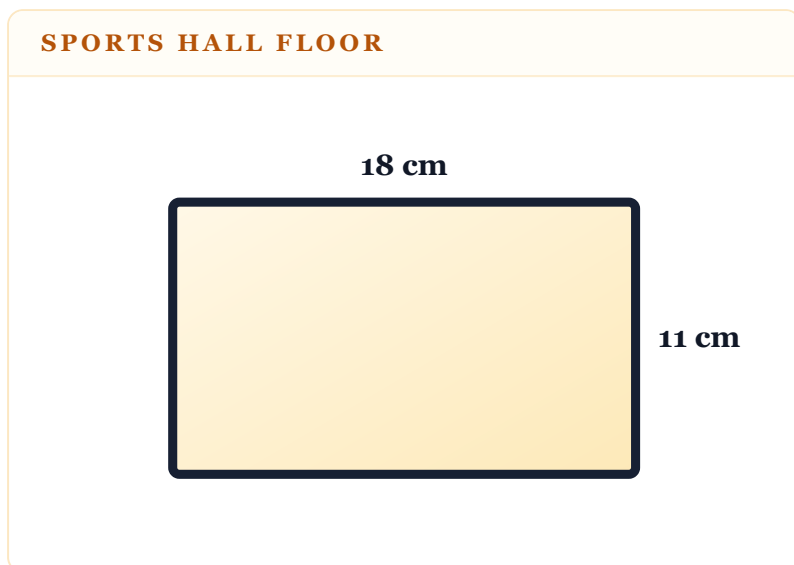
29 A triangle has one angle of 90° and another of 34° . What is the size of the third angle? [1]

- | | |
|--|--|
| ☐ A 56°
☐ C 50° | ☐ B 46°
☐ D 60° |
|--|--|

- 30 The diagram shows an L-shaped garden. The outer width is 12 m and outer height is 8 m, with a 5 m by 3 m rectangle cut from the bottom-right corner. What is the perimeter of the garden? [1]



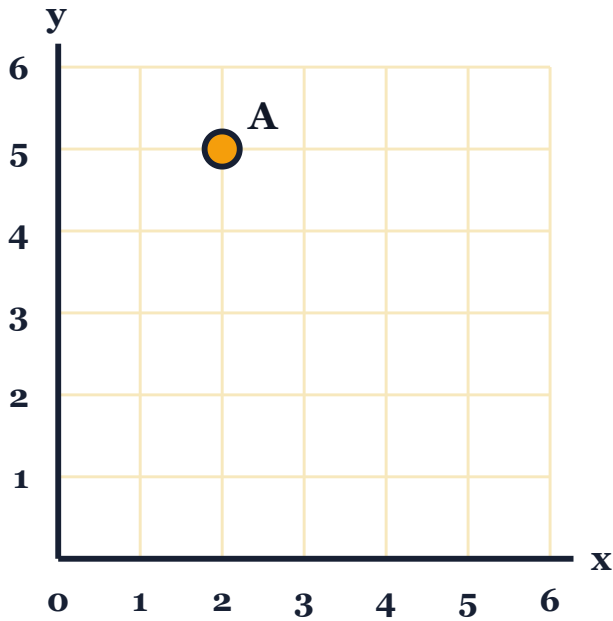
- (A) 40 m (B) 38 m
(C) 42 m (D) 46 m
- 31 A rectangular sports hall floor measures 18 m by 11 m. What is its area? [1]



- (A) 198 m² (B) 180 m²
(C) 58 m² (D) 209 m²
- 32 Which of these is NOT a property of a cube? [1]
- (A) It has 5 faces (B) It has 6 faces
(C) It has 8 vertices (D) It has 12 edges
- 33 How many lines of symmetry does a regular hexagon have? [1]
- (A) 6 (B) 5
(C) 4 (D) 3

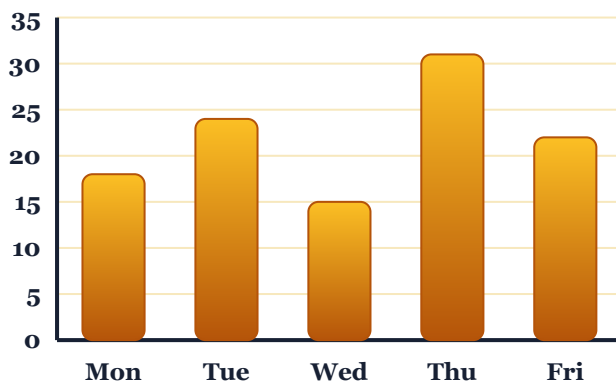
- 34 Point P is plotted on the grid at (2, 5). Point Q is 3 units to the right and 2 units down from P. What are the coordinates of Q? [1]

POINT P



- 35 The bar chart shows the number of books borrowed from the school library each day one week. How many more books were borrowed on Thursday than on Monday? [1]

BOOKS BORROWED THIS WEEK



- ☐ A 13 ☐ B 12
☐ C 9 ☐ D 16

- 36 The table shows how pupils in Year 5 travel to school. How many more pupils walk than travel by bus? [1]

HOW YEAR 5 TRAVEL TO SCHOOL

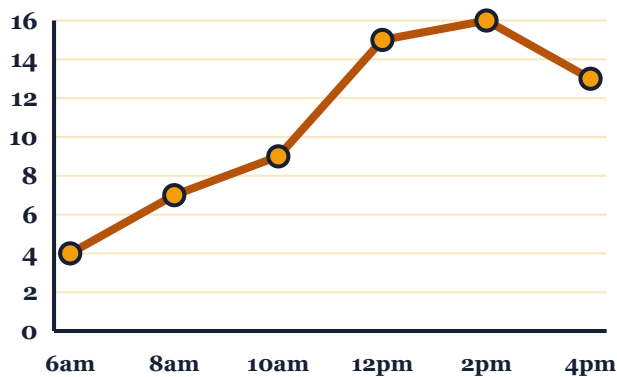
Method	Number of pupils
Walk	34
Car	21
Bus	19
Bicycle	16

- (A) 15 (B) 13
(C) 18 (D) 21
- 37 Five friends timed how long it took them to solve a puzzle: 8 minutes, 11 minutes, 6 minutes, 13 minutes and 12 minutes. What was their mean (average) time? [1]

- (A) 10 minutes (B) 11 minutes
(C) 9 minutes (D) 12 minutes

- 38 The line graph shows the temperature recorded at a weather station every 2 hours during one day. Between which two times did the temperature rise the most? [1]

WEATHER STATION TEMPERATURE



- (A) 10am and 12pm (B) 6am and 8am
(C) 12pm and 2pm (D) 2pm and 4pm

- 39 A school trip coach leaves at 09:45 and arrives 2 hours and 50 minutes later. At what time does it arrive? [1]

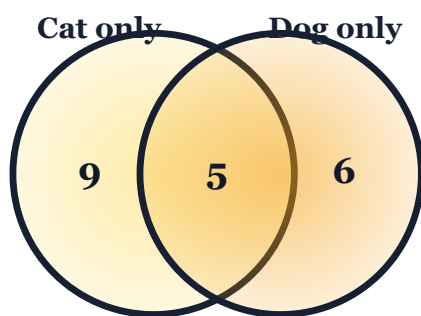
DEPARTURE TIME



- ☐ A 12:35 ☐ B 12:25
☐ C 11:35 ☐ D 12:45

- 40 In a class of 30 pupils, 9 have a pet cat only, 6 have a pet dog only, and 5 have both a cat and a dog. How many pupils have neither a cat nor a dog? [1]

PETS IN THE CLASS



- ☐ A 10 ☐ B 8
☐ C 11 ☐ D 14

Working